Agentic Engineer Capstone — Stakeholder One-Pager

What Problem Does This Solve?

Software changes often require implementation, review, testing, documentation, policy checks, and human approval. When handled manually or by one unrestricted AI agent, teams can face slower reviews, inconsistent decisions, unnecessary model cost, and higher risk from over-broad permissions.

This capstone demonstrates a governed engineering pipeline that divides work across specialized AI agents, deterministic automation, and human checkpoints.

What the System Does

The workflow coordinates specialized roles for implementation, code review, testing, project summaries, documentation, and workflow orchestration. Each role receives only the tools and data access required for its responsibility.

Stable structural work is moved out of AI agents and into deterministic code when possible. Human approval remains in the workflow for higher-risk or consequential actions.

Why This Matters to the Business

Metric	Before	After
Validation cycle time	45 seconds	0.2 seconds
Model-token cost	$0.003/run	$0/run
Review latency	~30 seconds	~5 seconds
Structural checks	7/7 passing	7/7 passing

The deterministic implementation also produced zero output variance across three repeated runs, showing that a stable workflow step can become faster, cheaper, and more predictable without reducing recorded validation quality.

Risk Reduction

The project identified governance near-misses including overly broad delete permissions, a read-only reviewer nearly receiving testing capability, and retrieval attempts above a role's approved data-classification level.

The final design addresses these risks through least-privilege tool access, role-specific MCP allow-lists, data-classification ceilings, read-only containers for advisory roles, policy and regression tests, human escalation, and red-team testing.

Evidence That Governance Works

An unauthorized documentation-writer attempt to call `write_entry` was denied by the storage MCP allow-list and recorded as `authorization_denied`, while the same role's permitted `read_entry` operation succeeded.

Final Production-Like Evidence

The final human-approved production-like workflow completed the expected **Planner** → **Implementer** → **Reviewer** → **Tester** path in **46.8 seconds**, using 33,519 of the bounded 38,000-token workflow budget.

The run used deterministic lexical retrieval with classification-ceiling enforcement, persistent-storage reads, an Implementer storage write, Reviewer verification, and Tester validation. Human approval was **approve** and completion was authorized.

Remaining Limitations

- The available evidence does not establish a trustworthy full-pipeline model-cost measurement, so no unsupported end-to-end cost claim is made.
- Human-checkpoint conditions are defined in governance policy; the successful final run recorded an explicit approval decision but did not require an escalation condition.
- One successful evaluated run is not sufficient evidence to claim a general 0% defect rate across all future workloads.

Next Steps

Continue measuring full-pipeline cost and quality across additional workloads, preserve governance checks in CI/CD, and use recorded audit and evaluation evidence to guide future right-sizing decisions.